

Math724 – Differential Equations II

Spring 2017

Final Exam

You must show all of your work to get a credit for a correct answer.

1. Let $U = \{(x, y) \in \mathbb{R}^2 \mid |x| \leq 1, 0 \leq y \leq 1\}$ and $\Gamma = \{x \in \mathbb{R} \mid |x| \leq 1\}$. Prove there exists a constant C , independent of u , such that for all $u \in H^1(U)$,

$$\|u(\cdot, 0)\|_{L^2(\Gamma)} \leq C\|u\|_{H^1(U)}.$$

2. Let $U = \mathbf{B}(\mathbf{0}, 1)$ be the unit disc in $\mathbb{R}^2$.
(a) Prove that for any u smooth up to the boundary of U

$$\|u\|_{L^2(\partial U)}^2 \leq C(\|u\|_{L^2(U)}^2 + \|u\|_{L^2(U)} \|\partial u / \partial r\|_{L^2(U)}).$$

- (b) Prove that the trace operator $T : H^1(U) \rightarrow L^2(\partial U)$ is compact. (*Hint: use $\|Tu\|_{L^2(\partial U)}^2 \leq C(\|u\|_{L^2(U)}^2 + \|u\|_{L^2(U)} \|\nabla u\|_{L^2(U)}$ that can be easily derived from (a).*)

3. Let $U \in \mathbb{R}^n$ be an open domain and suppose the operator L defined by

$$Lu = - \sum_{i,j=1}^n a^{i,j}(\mathbf{x}) u_{x_i x_j}$$

has smooth coefficients $a^{i,j}$ and is uniformly elliptic. Show that if $\lambda > 0$ is sufficiently large then whenever $Lu = 0$ we have

$$L(|Du|^2 + \lambda|u|^2) \leq 0.$$

4. Let $U \in \mathbb{R}^n$ be an open smooth bounded domain and $\partial U = \Gamma_1 \cup \Gamma_2$ where Γ_1 and Γ_2 each have positive surface area and smooth boundary. Consider the boundary value problem

$$\begin{cases} -\nabla \cdot (\mathbf{A}(\mathbf{x}) \nabla u) + c(\mathbf{x})u = f(\mathbf{x}), & \text{in } U, \\ u = 0, & \text{on } \Gamma_1, \\ (\mathbf{A}(\mathbf{x}) \nabla u) \cdot \mathbf{n} = 0, & \text{on } \Gamma_2, \end{cases}$$

where $c > 0$ and f are smooth, $\mathbf{n}$ is the outer unit normal on ∂U , and the $n \times n$ matrix-valued smooth function $\mathbf{A} = (a^{i,j})$ is symmetric and satisfies the uniformly ellipticity condition. Give a weak formulation of this problem and prove the existence of a weak solution in $H^1(U)$.

5. Let $U \in \mathbb{R}^n$ be an open smooth bounded domain with smooth boundary and define

$$\lambda = \inf \left\{ \int_U |Du|^2 d\mathbf{x} \mid u \in H_0^1(U) \text{ and } \int_U u^2 d\mathbf{x} = 1 \right\}.$$

Use energy method to show that, if u is a smooth solution of

$$\begin{cases} u_t - \Delta u = 0, & \text{in } U \times [0, \infty), \\ u = 0, & \text{on } \partial U \times [0, \infty), \end{cases}$$

then $\|u(\cdot, t)\|_{L^2(U)} \leq e^{-\lambda t} \|u(\cdot, 0)\|_{L^2(U)}$.